

PMF, CDF, Joint Distribution

CA2 @ EPS Course

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Instructions:

- Replace the placeholders (between `## TODO ##` and `## TODO ##`) with the appropriate details.
 - Ensure you run each cell after you've entered your solution.
-

Full Name: ...

SID: ...

The objective of this notebook is to understand the concepts of PMF and CDF and joint distribution.

Parameters

Generator(PCG64) at 0x2114A2A57E0

Simulation

Plot the RVs Distributions

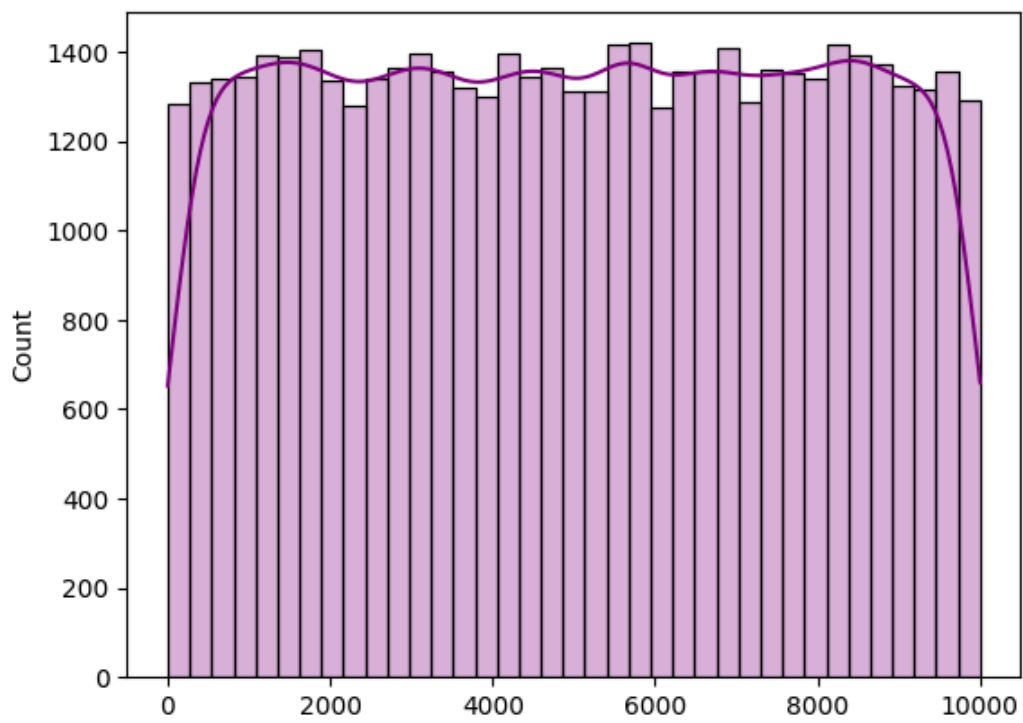
In this part I will plot the three distributions we've generated on three histograms.

Kernel Density Estimation (KDE) parameter shows the estimation of distribution on the current X axis. This means it is a smooth line which helps us to visualize and analyze the trends in our data. It is something like indicators in the stock charts in financial markets :)

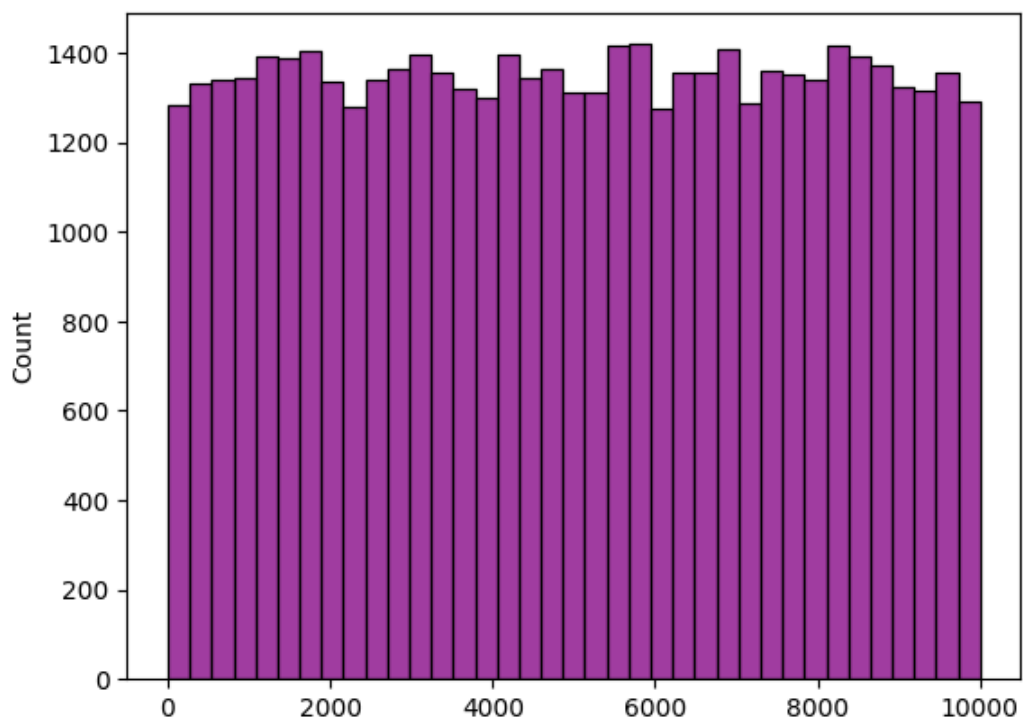
Actually KDE is an estimation for the Probability Density Function(PDF) according to the limited number of data we have.

I tried to change the color of kde in my plot but I couldn't. So I changed the transparency of the lines and increased the thickness of the kde lines.

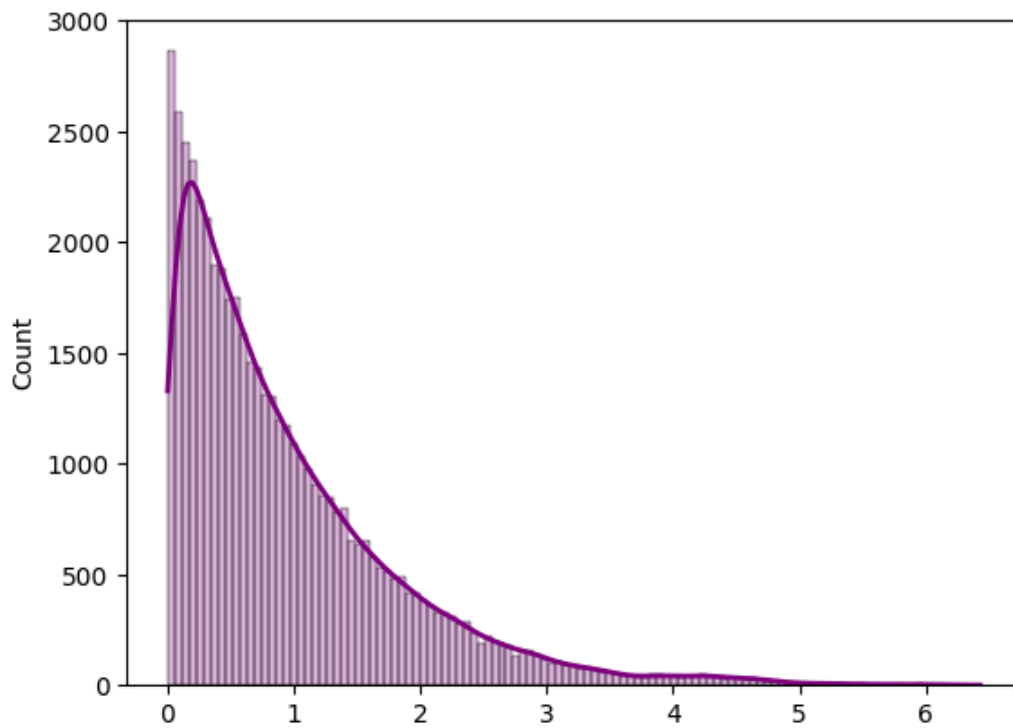
Also we can see that if we change the value of the kde argument to False, it won't plot the kde line. You can have this indicator on your histograms by putting the kde parameter equal to True.



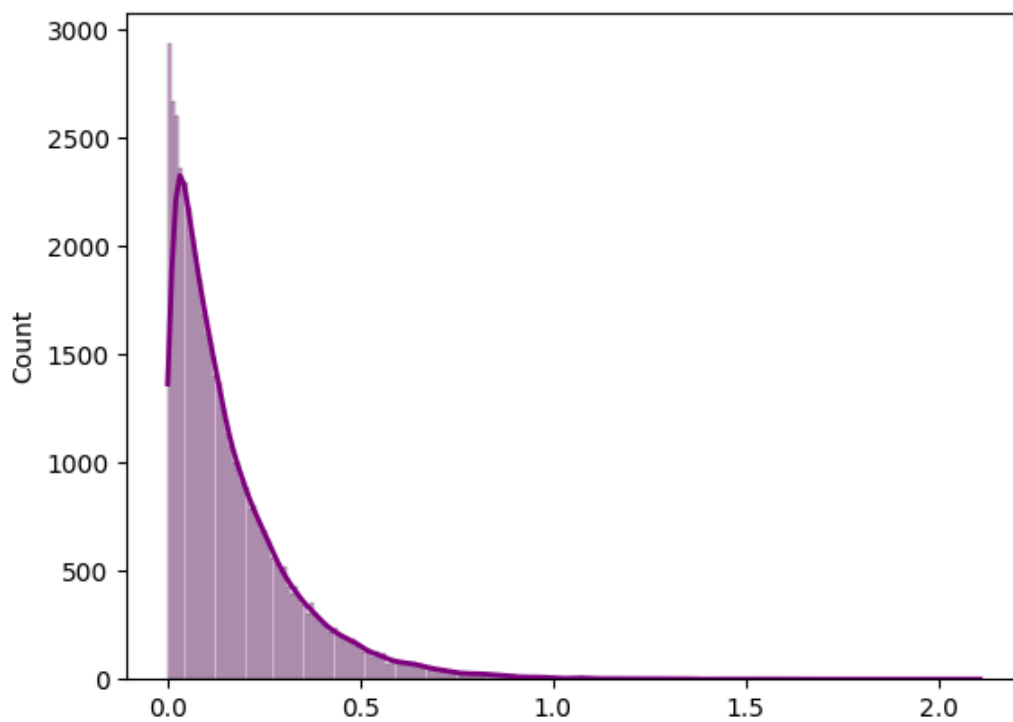
<Axes: ylabel='Count'>



<Axes: ylabel='Count'>

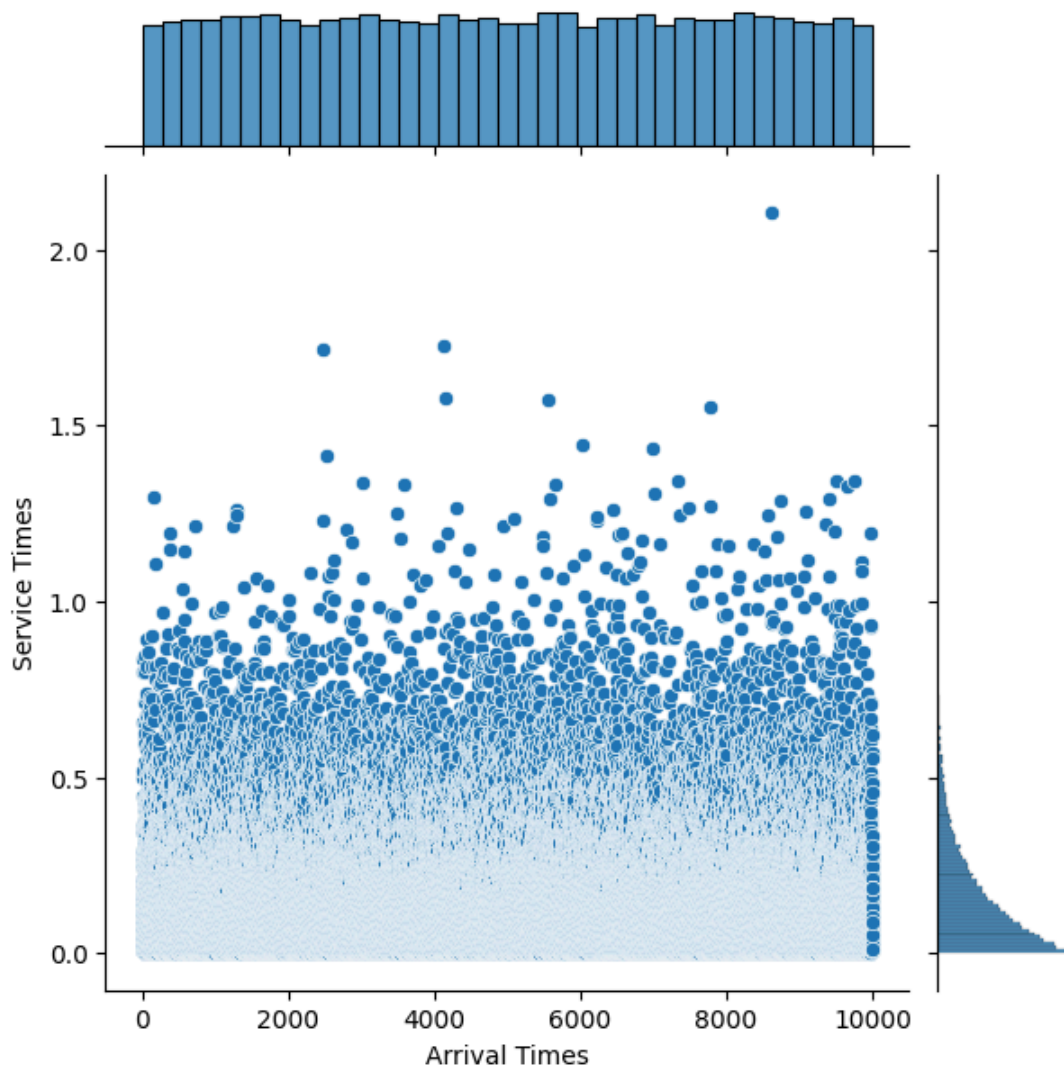


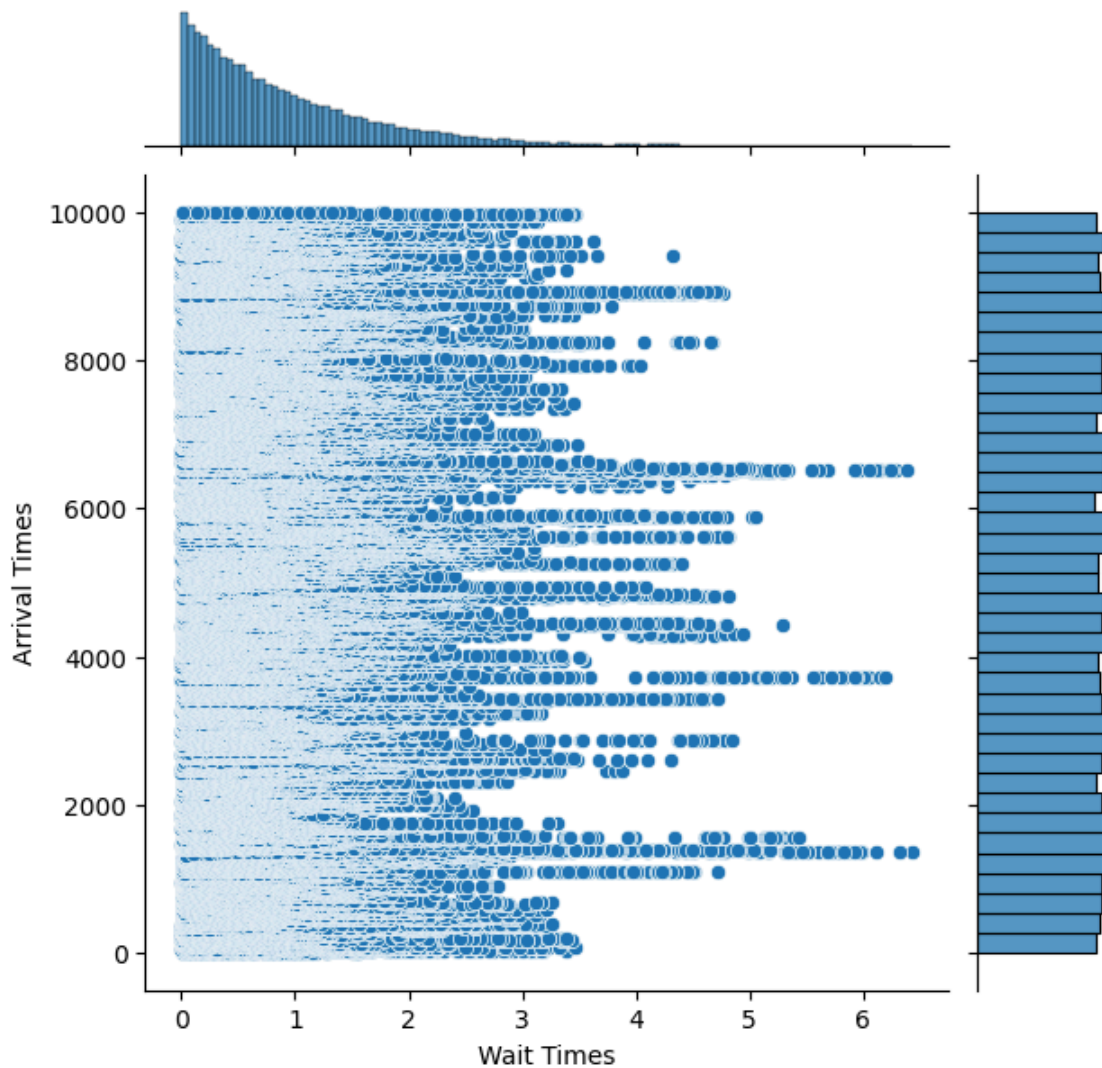
<Axes: ylabel='Count'>



Plot the Joint Distributions

Note: Use `sns.jointplot` to plot the joint distributions





Analysis of Joint Distributions

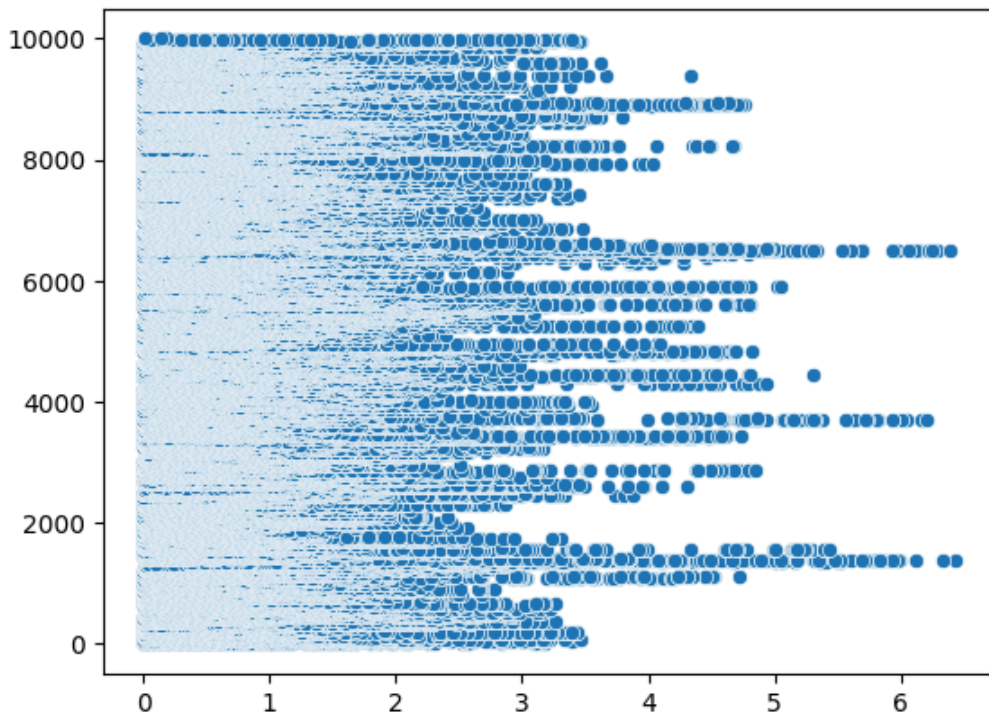
As we can see, the density of points is much more on the bottom of the plot. It's because the service time obeys the exponential distribution and it's much more probable to be around 0.

It's obvious that we can not fit a line on this data appropriately and the variables are absolutely not linearly dependent at all. This proves that the correlation coefficient for this data is too low(around 0.)

Scatter Plot and Correlation

-0.0009674242043743392

-0.0009674242043743502

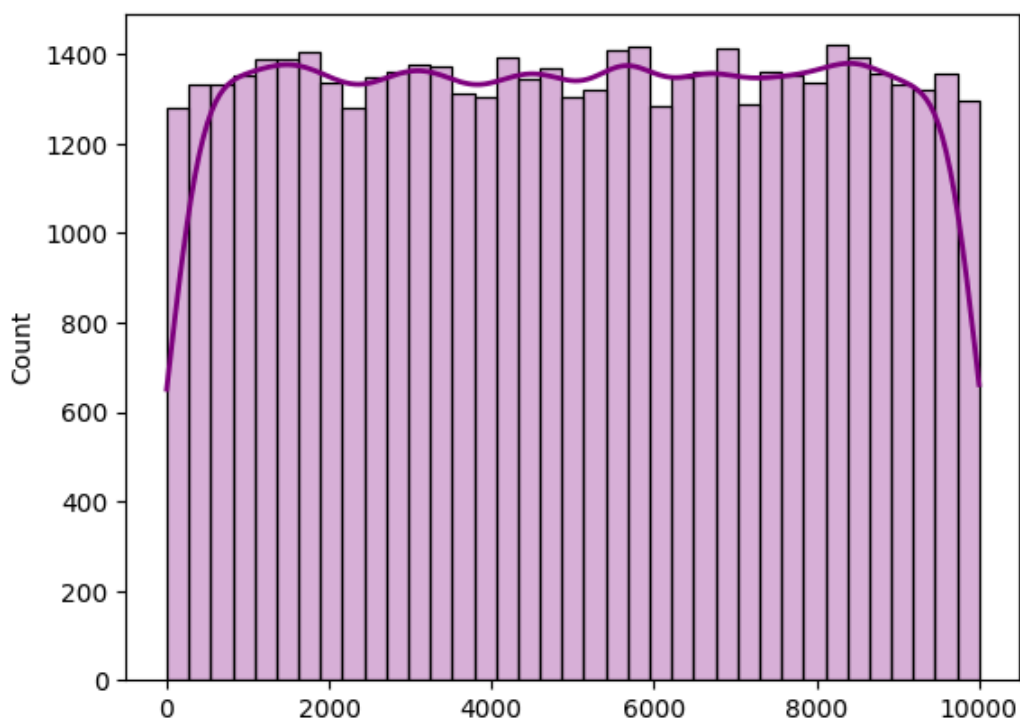


I both calculated the correlation coefficient using and without using `np.corrcoef()` function and they were equal(with a very slight difference.)

Calculating `total_times`

This list shows the times that each customer has left. To determine the distribution I have to plot the times and distinguish the distribution by that.

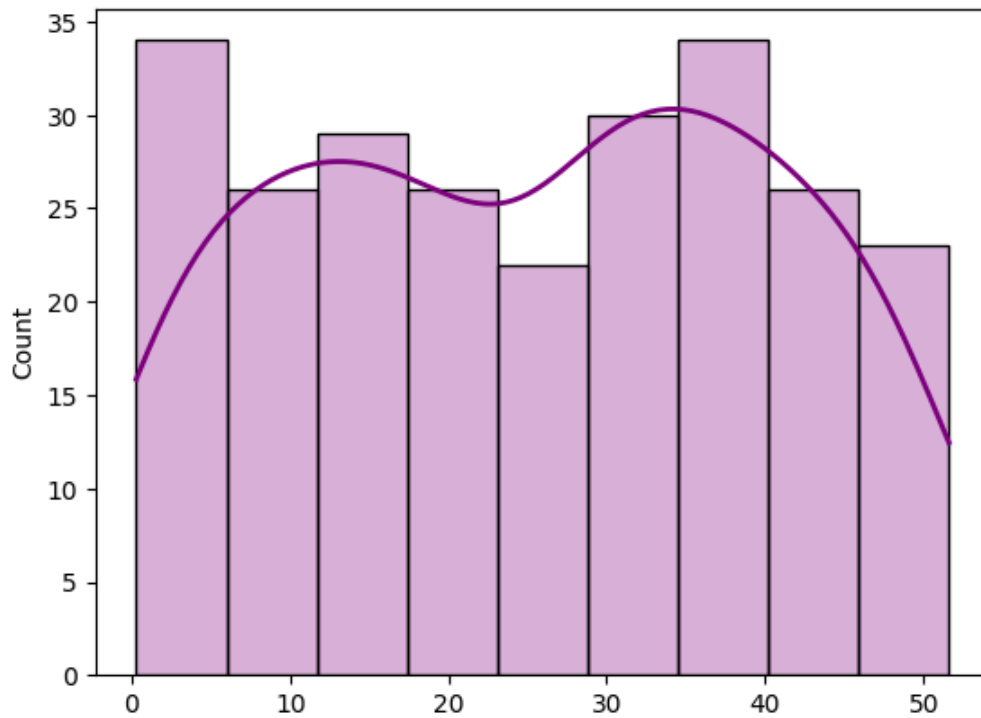
<Axes: ylabel='Count'>



According to the shape, the total time random variable distribution is uniform. This means that the time which a customer leaves the office follows a uniform distribution and has equal probabilities for each time in the period.

Conditional Wait Times

<Axes: ylabel='Count'>



According to the shape, this conditional distribution is uniform too.